SECTION - A (Marks 13)

Note: Selected options are based on the curriculum and question paper.

1. **Which shapes are used to represent concepts in a concept map?**
    - **Answer: B (Boxes & circles)**
2. **The smallest unit of memory in a digital computer:**
    - **Answer: A (Bit)**
3. **Which of the following is an example of primary data collection method?**
    - **Answer: C (Interviews)**
4. **Which is an example of AI very common?**
    - **Answer: D (Chat bot)**
5. **The tag used to define a hyperlink in HTML:**
    - **Answer: A (<a>)**
6. **Correct way to declare a variable in JavaScript:**
    - **Answer: D (var n=10;)**
7. **Attackers use phishing for:**
    - **Answer: B (Stealing data)**
8. **What is the purpose of malware?**
    - **Answer: B (Harm computers)**
9. **Which of the following datatype does JavaScript use to represent true or false values?**
    - **Answer: D (Boolean)**
10. **Which of the following is NOT one of the five phases of design thinking?**
    - **Answer: C (Analyze)**
11. **What is the correct sequence of IPO chart?**
    - **Answer: C (Input, process, output)**
12. **Cursor on a computer screen is controlled by:**
    - **Answer: C (Mouse)**
13. **Which mode of communication can flow data in one direction only?**
    - **Answer: A (Simplex)**

SECTION - B (Marks 22)

Q. 2 (i)

- **Purpose of Software Piracy:** The illegal copying, distribution, or use of software without permission from the creator, usually to avoid paying for it.
- **Example:** Downloading a "cracked" version of Microsoft Office or Adobe Photoshop from an unauthorized website.
- **OR: How Machine Learning assists in decision making:** It uses algorithms to analyze large datasets, identify patterns, and provide predictions or recommendations, helping humans make faster and more accurate data-driven choices.

Q. 2 (ii)

- **Differences between System and Application Software:**
    1. **Purpose:** System software manages computer hardware (e.g., Windows), while Application software performs specific tasks for users (e.g., MS Word).
    2. **Necessity:** System software is essential for the computer to run; Application software is optional and installed based on user needs.
- **OR: Differences between Qualitative and Quantitative Data:**
    1. **Qualitative:** Descriptive data (e.g., colors, textures, smells).
    2. **Quantitative:** Numerical data that can be measured (e.g., height, weight, age).

Q. 2 (iii)

- **How Cloud Computing allows online storage:** It uses remote servers hosted on the internet to store, manage, and process data instead of a local hard drive, allowing access from any device with internet.
- **OR: Rewrite JavaScript Code (Corrected):**

- **OR: Rewrite JavaScript Code (Corrected):**

JavaScript

```javascript
<script type="text/javascript">
document.write("Welcome");
</script>
```

(Errors fixed: 'Document' to 'document', 'writ' to 'write', added missing quote and backslash in closing tag).

Q. 2 (iv)

HTML

```html
<html>
<head>
<title>Main Page</title>
</head>
<body>
<p>Introduction about the page</p>
</body>
</html>
```

(Errors fixed: Tag order, fixed closing title tag, corrected closing paragraph slash).

- **OR: Why Operating System is important:**
    1. It acts as an interface between the user and the hardware.
    2. It manages resources like memory, CPU, and disk space.

Q. 2 (v)

- **Differences between Natural and Artificial System:**
    1. **Natural:** Exists in nature without human intervention (e.g., Solar system, Digestive system).
    2. **Artificial:** Man-made systems designed for a purpose (e.g., Computer system, Transport system).
- **OR: Differences between Static and Dynamic Website:**
    1. **Static:** Fixed content that remains the same for every visitor.
    2. **Dynamic:** Content changes based on user interaction or time (e.g., Facebook, News sites).

Q. 2 (vi)

- **Differences between Freeware and Shareware:**
    1. **Freeware:** Available for free for an unlimited time.
    2. **Shareware:** Provided for free on a trial basis, after which payment is required.
- **OR: Full Duplex vs. Half Duplex:**

1. **Half Duplex:** Data can flow both ways but not at the same time (e.g., Walkie-talkie).
2. **Full Duplex:** Data can flow both ways simultaneously (e.g., Telephone call).

Q. 2 (vii)

- **How Packet Switching works:** Data is broken into small chunks called "packets." Each packet is sent independently through the network and reassembled at the destination.
- **OR: Four Phases of Design Thinking:**
  1. Empathize 2. Define 3. Ideate 4. Prototype.

Q. 2 (viii)

- **Representation in Concept Map:** Concepts are represented as nodes (usually boxes or circles) and connected by labeled arrows to show relationships.
- **OR: Secondary Data Collection Examples:**
  1. Books/Journals 2. Government records 3. Online databases 4. Past research studies.

Q. 2 (ix)

- **Logical vs. Algorithmic Thinking:** Logical thinking is the ability to analyze a situation and come up with a sensible solution, whereas algorithmic thinking is the step-by-step process used to reach that solution.
- **OR: Output of JavaScript Code:** Admission granted *(Since 5 is greater than 3)*.

Q. 2 (x)

- **Importance of Flowchart:** It provides a visual representation of the logic of a problem, making it easier to understand, analyze, and convert into actual code.
- **OR: Purpose of Output Device:** To present data/information from the computer to the user (e.g., a Monitor displays images, a Printer produces hard copies).

Q. 2 (xi)

- **Value of x:** 7 *(Calculation: 10 - 3)*.
- **OR: Why Design Thinking is iterative:** It allows teams to continuously refine their ideas and prototypes based on feedback and testing results until the best solution is found.

SECTION - C (Marks 20)

Q. 3: BUS vs. STAR Topology

- **BUS Topology Features:** Uses a single central cable; hard to troubleshoot; if the main cable fails, the whole network goes down.
- **STAR Topology Features:** All devices connect to a central Hub/Switch; easy to add new devices; if one cable fails, only that device is affected.

Q. 4: Artificial Intelligence (AI)

- **Definition:** The simulation of human intelligence processes by machines, especially computer systems.
- **Three Areas:**
  1. **Robotics:** Programmable machines that can carry out actions.
  2. **Machine Learning:** Systems that learn from data.
  3. **Natural Language Processing (NLP):** Computers understanding human speech/text.

Q. 5: CPU Components

- **CPU:** The "Brain" of the computer that processes instructions.
- **ALU (Arithmetic Logic Unit):** Performs mathematical calculations ($+$, $-$, $\times$, $\div$) and logical comparisons ($<$, $>$, $=$).
- **CU (Control Unit):** Coordinates the flow of data between the CPU and other hardware; it fetches, decodes, and executes instructions.

Q. 6: Big Data

- **Purpose:** To analyze extremely large datasets to reveal patterns, trends, and associations.
- **Applications in Business:**
  1. **Customer Insights:** Understanding buying habits.
  2. **Market Trends:** Predicting future demand.
  3. **Risk Management:** Detecting fraud or financial risks.

Q. 6 (OR Part 1): Computing Innovation

- **Computing Innovation:** The creation and application of new computer technologies to solve problems or improve life.
- **Three Benefits:**
  1. Increased productivity.
  2. Global connectivity and communication.
  3. Improved healthcare through medical technology.

Q. 6 (OR Part 2): HTML Code

** This is a bold, italic, <u>underlined</u>, and emphasized paragraph. **

Q. 6 (OR Part 3): Flowchart (Sum & Product)

Q. 6 (OR Part 4): JavaScript (1 to 10)

JavaScript

```javascript
for (var i = 1; i <= 10; i++) {
  document.write(i + "<br>");
}
```

SECTION-A (Marks 13)

1. **Which one is a type of artificial system?**
    - **Answer:** B. Hospital system
2. **Which memory type is primary and volatile?**
    - **Answer:** B. RAM
3. **Which unit performs arithmetic and logic operations in a computer?**
    - **Answer:** B. ALU
4. **Which symbol represents processing?** (Note: Standard flowchart symbol is a rectangle).
    - **Answer:** B. (Rectangle/Box symbol)
5. **Main role of an algorithm in programming is:**
    - **Answer:** B. Step by step plan
6. **Which HTML tag is used for the main heading?**
    - **Answer:** B. <h1>
7. **What is the purpose of CSS in web development?**
    - **Answer:** C. Style the web page
8. **Which JavaScript keyword declares a variable?**
    - **Answer:** A. var
9. **Which component transmits data between parts of the computer?**
    - **Answer:** C. Bus
10. **Which one is an example of continuous data?**
    - **Answer:** C. Temperature
11. **How can AI be helpful in education?**
    - **Answer:** C. Global access
12. **Which one is NOT the type of malicious software?**
    - **Answer:** D. Information privacy
13. **Which one describes the main advantage of cloud computing?**
    - **Answer:** A. Access from anywhere

SECTION-B (Marks 22)

Q. 2 (i)

- **Natural System:** A system that occurs naturally in the universe without human intervention.
    - **Examples:** The Solar System, the Human Digestive System.
- **OR - Decomposition:** The process of breaking down a complex problem into smaller, more manageable parts.

Q. 2 (ii)

- **RAM vs. ROM:**
    1. **RAM** is volatile (loses data when power is off), whereas **ROM** is non-volatile.
    2. **RAM** is used for temporary storage of running programs, while **ROM** stores permanent instructions like BIOS.
- **OR - Computational Thinking:** It helps in problem-solving by using techniques like decomposition, pattern recognition, and algorithms to find efficient solutions.

Q. 2 (iii)

- **Input Devices:** Hardware components used to provide data and control signals to a computer.
    - **Examples:** Keyboard, Mouse.
- **OR - JavaScript Output:**
    - **Code:** document.write("Computer Science!")
    - **Output:** Computer Science!

Q. 2 (iv)

- **Control Unit (CU):** It acts as the "brain" of the CPU, directing the flow of data and coordinating the activities of all other computer components.
- **OR - Cloud Computing Uses:**
    1. Online data storage and backup (e.g., Google Drive).
    2. Hosting web applications and collaborative editing.

Q. 2 (v)

- **Data Bus vs. Address Bus:**
    1. **Data Bus** carries the actual data between the CPU and memory.
    2. **Address Bus** carries information about where the data is being sent or retrieved from.
- **OR - HTML Tags:**
    - <b>: Used to make text **bold**.
    - <u>: Used to underline text.

Q. 2 (vi)

- **Ports:** Interface points on a computer where external devices can be connected.
    - **Examples:** USB Port, HDMI Port.
- **OR - Information Privacy:** The purpose is to protect sensitive personal data from unauthorized access, ensuring confidentiality and security.

Q. 2 (vii)

- **Design Thinking Phases:**
    1. Empathize (understand the user).
    2. Define (state the problem).
- **OR - Static vs. Dynamic Websites:**
    1. **Static:** Content is fixed and remains the same for every visitor.
    2. **Dynamic:** Content changes based on user interaction or data from a database.

Q. 2 (viii)

- **1st Generation Computer Features:**
    1. Used Vacuum Tubes as core components.
    2. Very large in size and consumed a lot of electricity.
- **OR - Correct HTML Code:**

HTML

```html
<html>
<head><title>main page</title></head>
<body>
<p>Introduction of computer</p>
</body>
</html>
```

Section (ix)

Main: Write a JavaScript statement to declare and initialize a variable.

In modern JavaScript, you typically use let or const (though var is also available).

JavaScript
let score = 100;

- **Declare:** let score (tells the computer to reserve memory).

- **Initialize:** = 100 (assigns the starting value).

OR: Briefly explain the concept of Von Neumann Architecture.

The **Von Neumann Architecture** is a theoretical design for a computer where the program data and the instruction data are stored in the same **shared memory**. It consists of a Central Processing Unit (CPU), a Control Unit, an Arithmetic Logic Unit (ALU), memory, and Input/Output interfaces.

Section (x)

Main: What is the purpose of operators in JavaScript?

Operators are special symbols used to perform operations on variables and values. Their primary purposes are:

1. **Arithmetic:** Performing math (e.g., +, -, *).
2. **Comparison:** Comparing two values to return true/false (e.g., ==, >).
3. **Assignment:** Storing values into variables (e.g., =).

OR: Write down any two differences between discrete and continuous data.

Feature	Discrete Data	Continuous Data
Nature	Can only take specific, distinct values (integers).	Can take any value within a range (fractions/decimals).
Counting	Usually counted (e.g., number of students).	Usually measured (e.g., height, temperature).

Section (xi)

Main: Why is debugging important in programming?

Debugging is the process of finding and fixing errors (bugs) in software code. It is important because:

- It ensures the program runs correctly and produces the **expected output**.
- It improves the **security** and **stability** of the software by removing unexpected crashes or vulnerabilities.

OR: Briefly explain the concept of phishing.

Phishing is a type of cyberattack where scammers trick people into revealing sensitive information, such as passwords or credit card numbers. This is usually done by sending fraudulent emails or messages that appear to come from a legitimate source (like a bank or a popular website).

Q.3

Main: Explain half-duplex and full-duplex modes with one example each.

- **Half-Duplex:** Data can be transmitted in both directions, but **not at the same time**. While one device is sending, the other must wait and receive.
    - *Example:* A Walkie-Talkie.
- **Full-Duplex:** Data can be transmitted in both directions **simultaneously**. Both devices can send and receive at the same time.
    - *Example:* A Telephone call or a smartphone.

OR: Explain two principles of computational thinking with examples.

1. **Decomposition:** Breaking down a complex problem into smaller, more manageable parts.
    - *Example:* Breaking down the process of "making a cake" into smaller tasks like buying ingredients, mixing the batter, and baking.
2. **Abstraction:** Focusing on the important information and ignoring unnecessary details.
    - *Example:* When using a map, you focus on the roads and landmarks needed for directions, ignoring the specific types of trees or the colors of the houses.

Q.4

Main: Write an HTML code to create a webpage with heading, paragraph, and link.

HTML

```html
<!DOCTYPE html>
<html>
<head>
    <title>My Webpage</title>
</head>
<body>

    <h1>This is a Heading</h1>
    <p>This is a paragraph of text explaining the content of the page.</p>
    <a href="https://www.google.com">Click here to visit Google</a>

</body>
</html>
```

OR: Explain the differences between system software and application software with suitable examples.

- **System Software:** Designed to run a computer's hardware and application programs. It acts as a platform for other software.
    - *Examples:* Operating Systems (Windows, macOS, Linux), Device Drivers.
- **Application Software:** Designed to perform specific tasks for the user.
    - *Examples:* Microsoft Word (word processing), Google Chrome (web browsing), Photoshop (image editing).

Q.5

Main: Write a JavaScript program that prints numbers from 10 to 1.

JavaScript

```javascript
for (let i = 10; i >= 1; i--) {
    console.log(i);
}
```

OR: What is big data? Explain any three advantages of big data.

Big Data refers to extremely large data sets that are too complex to be dealt with by traditional data-processing software.

Advantages:

1. **Better Decision Making:** Businesses can analyze trends to make more informed choices.
2. **Cost Reduction:** Identifies efficient ways of doing business and reduces waste.
3. **Improved Customer Service:** Understanding customer preferences allows for personalized experiences.

Q.6

Main: What is Artificial Intelligence (AI)? Explain any three areas of AI.

AI is the simulation of human intelligence processes by machines, especially computer systems.

Areas of AI:

1. **Machine Learning:** Systems that learn and improve from experience without being explicitly programmed.
2. **Natural Language Processing (NLP):** Enabling computers to understand and process human language (e.g., Siri or Alexa).
3. **Robotics:** Designing and building robots to perform tasks that are usually done by humans.

OR: Draw a flowchart to find the area of a triangle.

To find the area of a triangle, the formula is: $Area = 0.5 \times base \times height$.

SECTION - A (Objective)

Q#	Question	Answer
1	What is correct sequence of IPO chart?	C. Input, process, output
2	Which mode of communication allows simultaneous sending and receiving?	D. Full duplex
3	Example of Government secondary data collection method?	A. Government records
4	Area of AI where facial recognition is used?	A. Computer vision
5	Tag used to define a hyperlink in HTML?	A. <a>
6	Keyword used to declare variables in JavaScript?	**C. var** (Note: const is also used, but var is the traditional answer)
7	Attackers use phishing for:	B. Stealing data
8	Which of the following is an example of spyware?	A. Online tracking
9	Which is NOT a valid JavaScript data type?	A. Long
10	Type of process in design thinking?	B. Iterative and non-linear
11	First step in solving a simple problem?	C. Define and analyze the problem
12	Identify the output device:	B. Printer
13	Smallest unit of memory in a digital computer:	A. Bit

SECTION - B (Short Questions)

i. Purpose of Phishing & Example

- **Purpose:** To steal sensitive data like passwords, credit card numbers, or login credentials by disguised as a trustworthy entity.
- **Example:** Receiving an email that looks like it's from your bank asking you to click a link and "verify" your account details.
- **OR: Two Positive Impacts of AI**
    1. **Automation:** Performing repetitive tasks quickly and accurately.
    2. **Healthcare:** Assisting in early disease diagnosis and personalized medicine.

ii. 2nd vs 3rd Generation Computers

1. **Technology:** 2nd Gen used Transistors; 3rd Gen used Integrated Circuits (ICs).
2. **Size/Speed:** 3rd Gen computers were smaller and faster than 2nd Gen.

- **OR: Ordinal vs Nominal Data**

1. **Ordinal:** Data that has a natural rank or order (e.g., Exam grades A, B, C).
2. **Nominal:** Data used for labeling variables without quantitative value (e.g., Hair color, Gender).

iii. Correct JavaScript Code

- **Corrected Code:**

JavaScript

```javascript
<script type="text/javascript">
  document.write("Hello world");
</script>
```

- **OR: Purpose of Input Device**
    - **Keyboard:** Used to input textual data and commands into the computer.

iv. Correct HTML Errors

- **Corrected Code:**

```html
<html>
  <head>
    <title>Main Page</title>
  </head>
  <body>
    <h2>Main heading</h2>
  </body>
</html>
```

- **OR: Asynchronous vs Synchronous**
    1. **Asynchronous:** Sends data in character-by-character flow with start/stop bits.
    2. **Synchronous:** Sends data in blocks or frames synchronized by a clock signal.

v. Natural vs Artificial System

1. **Natural:** Occurs in nature without human intervention (e.g., Solar system).
2. **Artificial:** Man-made systems designed for specific tasks (e.g., A car, a computer).

- **OR: Number vs Boolean in JS**

1. **Number:** Represents numeric values (e.g., 10, 3.14).
2. **Boolean:** Represents only two logical values: true or false.

vi. Spam vs Spyware

1. **Spam:** Unsolicited bulk messages sent via email.
2. **Spyware:** Malicious software that secretly monitors and collects user data.

- **OR: Simplex vs Half Duplex**

1. **Simplex:** Unidirectional communication (e.g., Television broadcast).
2. **Half Duplex:** Bi-directional but not at the same time (e.g., Walkie-talkie).

vii. Circuit Switching

- It establishes a dedicated physical path between two nodes for the duration of the communication session.
- **OR: Four Phases of Design Thinking**
    1. Empathize, 2. Define, 3. Ideate, 4. Prototype.

viii. Concepts in Concept Map

- Concepts are represented in circles or boxes (nodes) and connected by labeled arrows to show relationships.
- **OR: Four Examples of Primary Data Collection**
    1. Surveys, 2. Interviews, 3. Observations, 4. Experiments.

ix. Role of Computational Thinking

- It helps break complex problems into smaller, manageable parts (decomposition) and identifies patterns to find efficient solutions.
- **OR: JavaScript Output**
    - **Output:** You have Passed (Since 45 > 40).

x. Importance of Flowchart

- It provides a visual representation of the logic and steps required to solve a problem, making it easier to understand and debug.
- **OR: Cloud Computing Storage**
    - Users upload data to remote servers managed by providers, accessing it via the internet from any device.

xi. Value of y

- `z = 10 + 10 = 20`; `y = 20 - 5 = 15`. **Value of y is 15.**
- **OR: Design Thinking & Innovation**
    1. Focuses on human needs to find relevant solutions.
    2. Encourages iterative prototyping to refine and improve ideas.

SECTION - C (Long Questions)

Q. 3: Ring vs Star Topology

Feature	Ring Topology	Star Topology
Structure	Each device connects to two neighbors in a circle.	All devices connect to a central hub/switch.
Failure	One cable failure can take down the whole network.	One cable failure only affects that specific device.
Performance	Performance slows as more nodes are added.	Easier to expand without affecting performance significantly.

Q. 4: Computing Innovation & Harmful Effects

- **Definition:** The use of computing technologies to create new products or processes that solve problems.
- **Harmful Effects:**
    1. **Privacy Concerns:** Increased data collection leads to loss of personal privacy.
    2. **Job Displacement:** Automation may replace human labor in various sectors.
    3. **Digital Divide:** Inequality between those who have access to technology and those who don't.
- **OR: System Software & Types**
    - **Definition:** Software designed to provide a platform for other software and manage hardware.
    - **Types:** 1. Operating Systems (e.g., Windows), 2. Utility Programs (e.g., Antivirus).

Q. 5: JS Program (1-20)

```
<script>
  for (var i = 1; i <= 20; i++) {
    document.write(i + "<br>");
  }
</script>
```

OR: Big Data Purpose & Benefits

- **Purpose:** To analyze massive volumes of data to find patterns and trends.
- **Benefits:** 1. Better decision-making, 2. Improved customer insights, 3. Increased operational efficiency.

Q. 6: Flowchart for Product and Average

1. Start -> 2. Input A, B, C, D -> 3. Calculate $Product = A \times B \times C \times D$ -> 4. Calculate $Average = (A + B + C + D)/4$ -> 5. Print Product, Average -> 6. Stop.

```
<p>This is a <b>bold</b> paragraph.</p>
<p>This is <i>italic</i> and <em>emphasized</em> text.</p>
<p>This text is <u>underlined</u>.</p>
```

Feature console.log() document.write()

📊 Comparison Table

Feature	`console.log()`	`document.write()`
Purpose	Used for **debugging** and testing code	Used to **display output on the webpage**
Where output appears	In the **browser console** (Inspect → Console tab)	Directly on the **webpage (browser screen)**
Effect on page	Does **not affect page content**	Can **overwrite the whole page** if used after loading
Usage time	Used during **development/debugging**	Rarely used in modern development
User visibility	❌ Not visible to normal users	✅ Visible to users
Best for	Checking variables, errors, logic	Simple demonstrations (not recommended in real apps)
Modern practice	✅ Widely used	❌ Considered outdated